



HACKTHEBOX

Jingle Bell Writeup



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Machine Author: CyberJunkie

Difficulty: **Easy**

Scenario

Torrin is suspected to be an insider threat in Forela. He is believed to have leaked some data and removed certain applications from their workstation. They managed to bypass some controls and installed unauthorised software. Despite the forensic team's efforts, no evidence of data leakage was found. As a senior incident responder, you have been tasked with investigating the incident to determine the conversation between the two parties involved.

Artifacts Provided

- jinglebell.zip -

4d857dd670e8f91479e201cd564dacc6480384c323735309f4ad8c0ab2a97ad9

```
[vivisghost@HackTheBox]~[~/Jingle_Bell]  
$sha256sum jinglebell.zip  
4d857dd670e8f91479e201cd564dacc6480384c323735309f4ad8c0ab2a97ad9  jinglebell.zip
```

Skills Learnt

- Extracting forensic evidence from Windows Notifications
- Analyzing SQLite databases
- Investigating insider threats

Tags

- Windows Forensics
- Incident Response

Initial Analysis

We begin by looking at what we have been given.

```
[vivisghost@HackTheBox]--[~/Jingle_Bell/Torrincase]
$tree -la
.
├── C
│   ├── Users
│   │   ├── Appdata
│   │   │   ├── Local
│   │   │   │   ├── Microsoft
│   │   │   │   │   ├── Windows
│   │   │   │   │   │   ├── Notifications
│   │   │   │   │   │   │   ├── wpndatabase.db
│   │   │   │   │   │   │   ├── wpndatabase.db-shm
│   │   │   │   │   │   │   ├── wpndatabase.db-wal
│   │   │   │   │   │   │   └── wpnidm
└── 9 directories, 3 files
```

We have 3 files, all components of a SQLite database.

1. The .db file holds the core database.
2. The .db-shm file manages shared access to the database for concurrency.
3. The .db-wal file temporarily logs changes before they are committed to the main database.

To examine the data, we used SQLitebrowser. (The .shm and .wal files will automatically be utilized if they're in the same directory.)

```
[vivisghost@HackTheBox]--[~/Jingle_Bell/Torrincase]
$sqlitebrowser C:/Users/Appdata/Local/Microsoft/Windows/Notifications/wpndatabase.db
```

Upon initial review, the relevant information appears to reside within the Notification table.

DB Browser for SQLite - C:/Users/Appdata/Local/Microsoft/Windows/Notifications/wpndatabase.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

Table: Notification

Order	Id	HandlerId	ActivityId	Type	Payload	Tag	Gr
1	26	31	121	BLOB	tile	<tile>...	Logo
2	27	32	121	BLOB	tile	<tile>...	Community
3	42	203	75	BLOB	tile	<tile>...	
4	43	204	75	BLOB	tile	<tile>...	
5	44	205	75	BLOB	tile	<tile>...	
6	45	206	75	BLOB	tile	<tile>...	
7	47	264	139	BLOB	tile	<?xml version="1.0" encoding="utf-8"?><tile><visual ...	
8	49	266	76	BLOB	tile	<tile><visual lang="en-US"><binding template="TileSmall" ...	
9	51	268	164	BLOB	toast	<toast activationType="protocol" launch="slack://channel?...	
10	52	269	129	BLOB	tile	<tile>...	
11	53	270	129	BLOB	tile	<tile>...	
12	54	271	139	BLOB	tile	<?xml version="1.0" encoding="utf-8"?><tile><visual ...	
13	55	272	139	BLOB	tile	<?xml version="1.0" encoding="utf-8"?><tile><visual ...	
14	56	273	129	BLOB	tile	<tile>...	
15	57	274	129	BLOB	tile	<tile>...	
16	58	275	164	BLOB	toast	<toast activationType="protocol" launch="slack://channel?...	
17	59	276	164	BLOB	toast	<toast activationType="protocol" launch="slack://channel?...	
18	60	277	164	BLOB	toast	<toast activationType="protocol" launch="slack://channel?...	
19	61	278	164	BLOB	toast	<toast activationType="protocol" launch="slack://channel?...	

Go to: 1

Editing row=9, column=6
Type: Text / Numeric; Size: 701 character(s)

Remote

Identity Select an identity to connect

DBHub.io Local Current Database

Name Last modified Size Commit

SQL Log Plot DB Schema Remote

UTF-8

The entries in the raw format were challenging to interpret. Using the Execute SQL tab we created a query to filter the data and exported it to a CSV for easier analysis.

DB Browser for SQLite - C:/Users/Appdata/Local/Microsoft/Windows/Notifications/wpndatabase.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

SQL* x

```
1 SELECT Payload FROM Notification WHERE Type = 'toast';
2
```

Payload
<toast activationType="protocol" ...
<toast activationType="protocol" ...
<toast activationType="protocol" ...
<toast activationType="protocol" ...
<toast activationType="protocol" ...
<toast activationType="protocol" ...

Execution finished without errors.
Result: 8 rows returned in 6ms
At line 1:
SELECT Payload FROM Notification WHERE Type = 'toast';

Editing row=4, column=0
Type: Text / Numeric; Size: 835 character(s)

Remote

Identity Select an identity to connect

DBHub.io Local Current Database

Name Last modified Size Commit

SQL Log Plot DB Schema Remote

UTF-8

By exporting the data as a CSV and using `sed` to insert newlines between entries, we have created a more readable format.

```
[vivisghost@HackTheBox]--[~/Jingle_Bell]
$cat Torrin_Output.csv | sed 's#</toast>#</toast>\n#g'
Payload
"<toast activationType="protocol" launch="slack://channel?id=D05422YSHGD&message=1681985348.xxx2&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message from Slackbot</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">What do you know, it works?</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/3d54bfdbacb9dfa162dbd82c15b81751.png"/></binding></visual><audio silent="true"/></toast>
"
"<toast activationType="protocol" launch="slack://channel?id=D0544UUC4UB&message=1681985806.920359&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message from cyberjunkie</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev accepted your invitation to join Slack – take a second to say hello.</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/35f6a85490effd9267c1e097b456bc77.png"/></binding></visual><audio silent="true"/></toast>
"
```

This will be enough for our initial analysis. Next, we need to read the content of the Slack messages to answer the questions.

Questions

1. Which software/application did Torrin use to leak Forela's secrets?

Launch `slack:channel` and `invitation to join Slack` are good indications of the application.

Answer: Slack

2. What's the name of the rival company to which Torrin leaked the data?

```
"<toast activationType="protocol" launch="slack://channel?id=D0544UUC4UB&message=1681985806.920359&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message from cyberjunkie</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev accepted your invitation to join Slack – take a second to say hello.</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/35f6a85490effd9267c1e097b456bc77.png"/></binding></visual><audio silent="true"/></toast>
```

Answer: PrimeTech Innovations

3. What is the username of the person from the competitor organization whom Torrin shared information with?

Answer found in same Notification as question #2.

Answer: Cyberjunkie-PrimeTechDev

4. What's the channel name in which they conversed with each other?

```
"<toast activationType="protocol" launch="slack://channel?id=C05451QSQM8&message=168198665.563319&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message in #forela-secrets-leak</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev: Just to confirm as we dont want forela's IT team to get suspicious Password for the archive server is :<br>"Tobdaf8Qip$re@1"</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/5ad0b5f5ad7976cea80bb0ae6af2cebf.png"/></binding></visual><audio silent="true"/></toast>
```

```
"<toast activationType="protocol" launch="slack://channel?id=C05451QSQM8&message=1681986724.763179&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message in #forela-secrets-leak</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev: Confirmation that password is "Tobdaf8Qip$re@1"</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/5ad0b5f5ad7976cea80bb0ae6af2cebf.png"/></binding></visual><audio silent="true"/></toast>
```

Answer: forela-secrets-leak

5. What was the password for the archive server?

Answer found in same Notification as question #4.

Answer: Tobdaf8Qip\$re@1

6. What was the URL provided to Torrin to upload stolen data to?

```
"<toast activationType="protocol" launch="slack://channel?id=C05451QSQM8&message=1681986889.660179&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message in #forela-secrets-leak</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev: https://drive.google.com/drive/folders/1vW97VBmxDZUIEuEUG64g5DLZvFP-Pd1l?usp=sharing, remember to upload the documents and pdfs too</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/5ad0b5f5ad7976cea80bb0ae6af2cebf.png"/></binding></visual><audio silent="true"/></toast>
```

Answer: <https://drive.google.com/drive/folders/1vW97VBmxDZUIEuEUG64g5DLZvFP-Pdll?usp=sharing>

7. When was the above link shared with Torrin?

```
"<toast activationType="protocol" launch="slack://channel?id=C05451QSQM8&message=1681986889.660179&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message in #forela-secrets-leak</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev: https://drive.google.com/drive/folders/1vW97VBmxDZUIEuEUG64g5DLZvFP-Pdll?usp=sharing, remember to upload the documents and pdfs too</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/5ad0b5f5ad7976cea80bb0ae6af2cebf.png"/></binding></visual><audio silent="true"/></toast>
```

Going to <https://www.epochconverter.com/> we can convert the time to UTC.

 **EpochConverter**

Epoch & Unix Timestamp Conversion Tools

The current Unix epoch time is **1732012942**

Convert epoch to human-readable date and vice versa

Timestamp to Human date [\[batch convert\]](#)

Supports Unix timestamps in seconds, milliseconds, microseconds and nanoseconds.

Assuming that this timestamp is in **seconds**:

GMT : Thursday, **April 20, 2023 10:34:49** AM

Answer: 2023-04-20 10:34:49

8. For how much money did Torrin leak Forela's secrets?

```
"<toast activationType="protocol" launch="slack://channel?id=C05451QSQM8&message=1681987020.043589&team=T054518ADUJ&origin=notification"><header id="T054518ADUJ" title="PrimeTech Innovations" activationType="protocol" arguments="slack://channel?team=T054518ADUJ"></header><visual><binding template="ToastGeneric"><text hint-wrap="false" hint-maxLines="1">New message in #forela-secrets-leak</text><text hint-maxLines="10" hint-style="bodySubtle" hint-wrap="true">Cyberjunkie-PrimeTechDev: Bank Account Number: 03135905179789  
Sent 10,000 £ to the above account as promised, cheers</text><image placement="appLogoOverride" hint-crop="circle" src="C:/Users/CYBERJ~1/AppData/Local/Temp/Notification Cache/5ad0b5f5ad7976cea80bb0ae6af2cebf.png"/></binding></visual><audio silent="true"/></toast>
```

Answer: £10000

